MILITARY SPECIFICATION

ADAPTOMETER, RADIOACTIVE PLAQUE, NIGHT VISION

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a portable apparatus designed for use as a night vision tester.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-L-31 — Laminated Thermosetting Sheets, Rods, and Tubes.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Box, Fiberboard.

PPP-B-676 — Boxes, Set-Up, Paperboard.

MILITARY

MIL-L-3891 — Luminescent Material and Equipment, (Nonradioactive).

MIL-L-10547 — Liners, Case, Waterproof.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

DRAWING

MILITARY MEDICAL SUPPLY AGENCY

1863 Adaptometer, Radioactive Plaque, Night Vision.

(Copies of specifications, standards, and drawing required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6515

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3. REQUIREMENTS

3.1 Material. Material shall be as hereinafter specified, and shall be of the highest quality throughout, suitable in all respects for the purpose intended.

3.2 Style, design, and dimensions. The adaptometer shall be an apparatus suitable for use as a night vision tester, consisting essentially of a radium plaque mounted in a plastic case, complete with instructions for use. Style, design, and dimensions shall be in accordance with Drawing 1863 and figure 1.

3.2.1 *Radium plaque.* The radium plaque shall consist of a radioluminescent material, using radium or carbon (isotope) 14 as the activator and blue emitting zinc sulfide as the phosphor, and shall be constructed as shown in drawing with a dark Landolt ring (not luminescent) centrally located on one side of plaque. Polonium, or other short lived activators, shall not be used. The background brightness of the plaque (without external excitation) shall be 0.12 ± 0.02 photopic microlamberts, when tested as specified in 4.4.1, and there shall be no change in brightness greater than ± 10 percent in 5 years. The radium content of the radioluminescent material for the plaque, if radium is used, shall be less than 4.5 micrograms. The luminous compound shall not shift or show cracks when plaque is vibrated. Fixing of the luminous compound shall be accomplished by applying it in mixture with low resin content (1-5 percent lacquer). Plaque shall be watertight, and in addition, edges of the plaque shall be sealed with a strip of waterproof pressure sensitive tape. Plaques shall be mounted in case (see 3.3) so that open side of Landolt ring is toward the right when instrument is viewed with slide open to the bottom. Plaque shall be firmly fixed within case so that position does not change in normal handling. The dark Landolt ring shall be clearly visible when the plaque is viewed in complete darkness.

3.3 Case. Case containing the radium plaque shall be fabricated of black bakelite in accordance with type I, grade xx of Specification L–L–31, and shall be constructed as shown in drawing.

3.4 Instruction plate. Each apparatus shall be furnished with an instruction plate permanently affixed to the apparatus with screws, as shown on drawing. The instruction plate shall be fabricated of frosted aluminum, with the instructions lithographed in black. The plate shall then be overcoated with clear baked on thermo-setting lacquer to protect the lithographing. The dimensions of the plate shall be as shown on drawing. The instructions shall be as follows:

RADIUM PLAQUE NIGHT VISION TESTER

FEDERAL STOCK NO. 6515–382–1000

INSTRUCTIONS FOR USE

A completely dark room must be provided for this test. A tent or building may be used at night in a location where outside illumination is minimal. To insure total darkness, the test room should be examined from the inside by a dark adapted observer and light leaks plugged.

The examinees must be dark adapted before taking the test. This may be accomplished either by having them sit in complete darkness or wear red goggles for 30 minutes. A short discussion of the nature of night vision and an explanation of scanning and other techniques of seeing at night should be given before the test. The necessity of looking "off-center" i.e., either above, below, or to the sides in order to see at night, must be emphasized over and over. Allow some practice in these methods before the actual test begins.

This tester may be used in either of two ways. It may be used to pick out individuals who have deficient night vision (those who cannot tell the position of the character at 5 feet) or it may be used to classify normal individuals with respect to the proficiency of

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their night vision. In the latter instance, the test is given at 5, 7, 9, and 11 feet, respectively. These distances are conviently measured by a cord looped at one end to go around the examinee's neck and knotted at appropriate distances.

A presentation consists of exposing the plaque to the subject with the break in the circle in one of four positions (left, right, up or down) for no longer than 10 seconds. If the position of the break is not called by the subject in this time, the trial is counted as an error. A random order of orientations should be used. A subject must name four out of the first four positions correctly to be considered as seeing the plaque at any given distance. If one or more of the first four presentations is missed, it is necessary to give ten presentations at that distance. The examinee then passes only if he can see eight out of ten.

A man who cannot see the position of the characters on the plaque at 5 feet has *unsatisfactory* night vision. Ability to see the characters at 5 feet but not at 7 feet is *below average;* at 7 feet but not at 9 feet is average; at 9 feet but not at 11 feet is *above average;* and at 11 feet is *superior.*

Experience demonstrates that approximately the following percentage distributions may be expected in a normal group of flyers.

Radium Plaque Score (character seen at)	Classification	Percent of Men in each Classification
Less than 5 feet ..	Unsatisfactory	2 percent
5 but not 7 feet ..	Below average	30 percent
7 but not 9 feet ..	Average	40 percent
9 but not 11 feet	Above average	26 percent
11 feet	Superior	2 percent

KEEP IN A COOL DRY PLACE
IF THIS PLAQUE HAS BEEN EXPOSED TO LIGHT, KEEP THE COVER CLOSED FOR 4 HOURS BEFORE USING.

3.5 Screws. All screws used in assembly of the apparatus shall be fabricated of brass with a black oxidized finish.

3.6 Serial numbers. Each unit shall bear a serial number positioned as shown on drawing. First number of sequence of serial numbers required for each contract will be supplied by Military Medical Supply Agency, 29th Street and 3rd Avenue, Brooklyn 32, New York.

3.7 Workmanship. Workmanship shall be first class throughout. Night vision tester shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.* When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

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4.2.1 *For examination.* Sampling for examination shall be conducted in accordance with Standard MIL–STD–105.

4.2.2 *For tests.* Sampling for tests shall be conducted in accordance with Standard MIL–STD–105. Requirement applicable to individual unit.

Component	Characteristic	Req.	Test procedure	Unit of product	Insp. level	AQL
Adaptometer plaque	Background brightness	3.2.1	4.4.1	1-plaque	L1*	**

* A minimum of two units shall be tested regardless of lot size.

** Acceptance number shall be zero for all sample sizes.

4.3 **Examination.** The adaptometers shall be examined to determine compliance with all requirements contained in this specification.

4.4 **Tests.** Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 *Background brightness of plaque.* The background brightness of the radium plaque shall be measured with the apparatus described in Specification MIL–L–3891 for low luminance measurements.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Unit of issue.* One (1) adaptometer, as specified, constitutes one unit of issue.

5.1.2 *Packaging quantities.* The number of units of issue indicated in table I shall be packaged in each unit, intermediate and exterior container as applicable for the required level of protection specified in the procurement document.

TABLE I—*Packaging quantities*

Unit package	Intermediate package	Exterior container
1 unit	Not required	12 units

5.1.2.1 *Packing variation permitted.* If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.* Each unit shall be packaged in a carton of appropriate size constructed in accordance with Specification PPP–B–636 or PPP–B–676. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to the contents. Each carton shall be adequately secured to prevent accidental opening.

5.2 Packing.

5.2.1 *Level A.*

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5.2.1.1 *Exterior container.* Exterior container shall be constructed in accordance with Specification PPP–B–621, table IV.

5.2.1.2 *Case liner.* Each level A box shall be lined with a waterproof case liner conforming to Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* Strapping, when required, shall be in accordance with applicable box specification.

NOTE: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.* Exterior container shall be constructed in accordance with Specification PPP–B–621, table II, or PPP–B–636, type I, class 1, fabricated from not less than 275-pound test fiberboard.

5.3 **Marking.**

5.3.1 *Interior package.* Each unit package shall be marked in accordance with section 5 of Standard MIL–STD–129.

5.3.2 *Exterior container.* Exterior container shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military Government requirements.

6.3 Transportation description. Transportation description applicable to this item is:

Optical Instrument not otherwise indexed by name.

Carload weight 10,000 pounds subject to Rule 34 of Uniform Freight Classification.

Truckload weight 10,000 pounds.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Army Medical Service
Navy—Bureau of Medicine and Surgery
Air Force

Preparing Activity:
Navy—Bureau of Medicine and Surgery
(MMSA)

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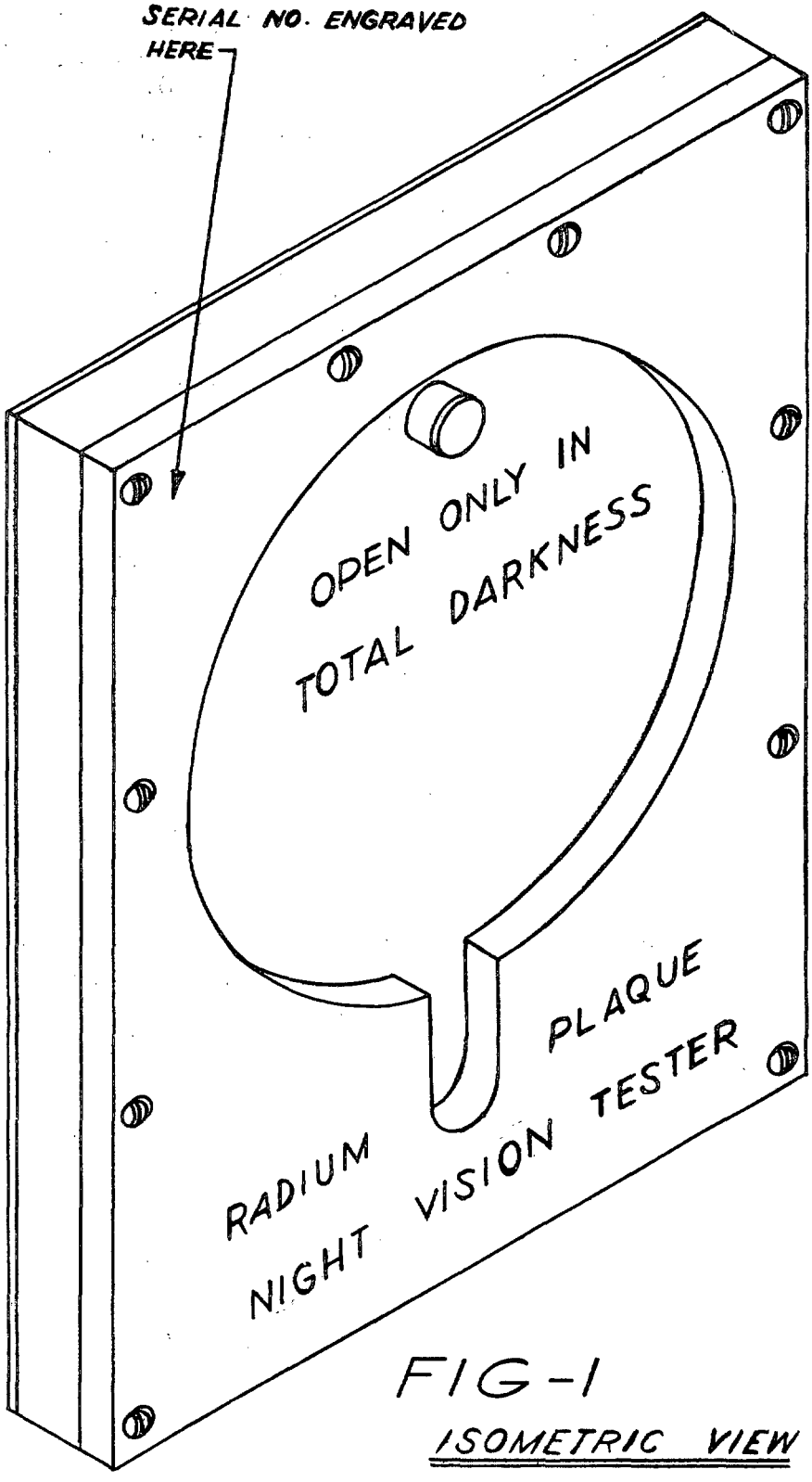


FIG-1

ISOMETRIC VIEW

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